



Open-ended Activity – Computer Networks

Radio Resource Management with Mist AI

Introduction:

Juniper Mist is a **cloud-based networking platform** that handles both wireless and wired network infrastructure from a single cloud console. Instead of relying on traditional on-premises controllers for network management, Mist shifts the control plane to the cloud. This “controller-less” model simplifies deployment and scaling: you connect access points and switches, and they “reach out” to the Mist cloud for configuration and management no bulky local controller hardware required.

At its core, Mist is powered by an AI engine named Marvis AI. This engine applies machine-learning, data-science and analytics techniques to monitor the network, make decisions, automate routine tasks, and optimize performance. In effect, Mist is a “cloud , AI and hardware” solution: enterprise-grade access points, switches, and other networking devices work together with cloud intelligence to deliver better performance, simplified operations, and more reliable connectivity.

Architecture of Mist:

The Mist cloud is built on a **microservices-based architecture**, meaning different functions (device management, analytics, configuration, data collection, security, etc.) run independently. This design makes the platform more flexible, scalable, resilient, and easier to update compared to traditional monolithic controller-based systems.

Because the control plane is cloud-native and centralized, you don’t need to manage hardware controllers on-site which simplifies operations, especially

when you have many access points or multiple sites. Additionally, the platform supports secure authentication, access control, and segmentation for wired, wireless, and IoT devices. Through its cloud services, Mist can maintain compliance, secure data paths, and manage user/device policies uniformly across network infrastructure.

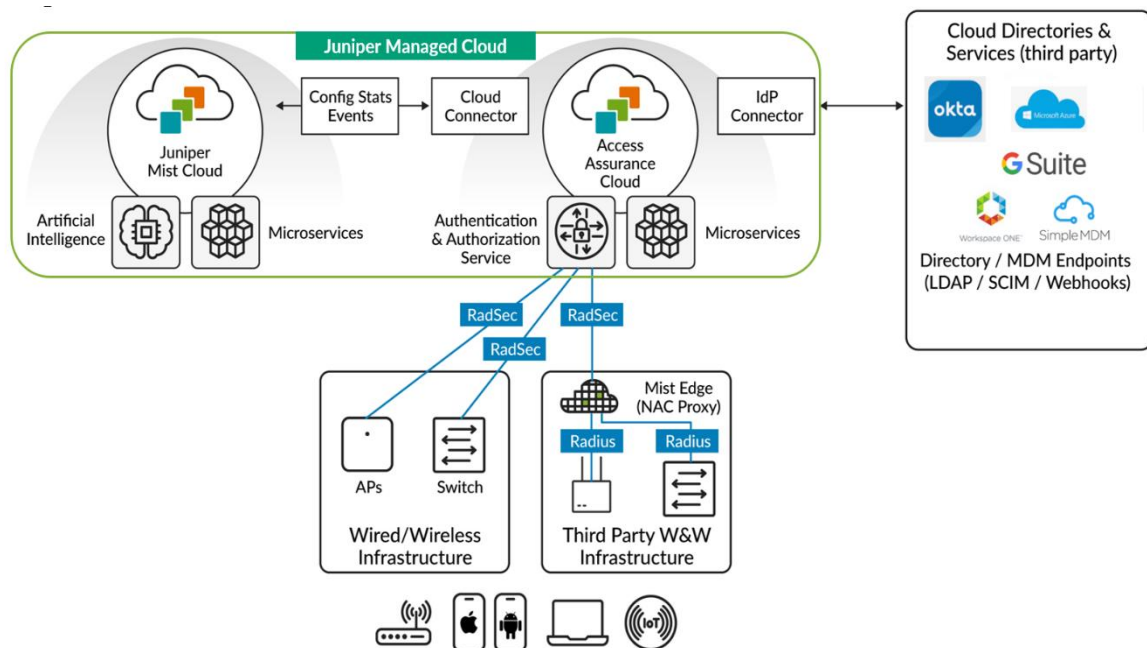


Figure 1 : Represent Mist Architecture

Main Features & Capabilities

Managed Wireless & Wired Connectivity (with Modern Hardware)

The company, known as Mist, provides Wi-Fi access points (APs), as well as Wi-Fi-enabled switches, which can vary from basic to more modern Wi-Fi technology, among other variants. It also supports Bluetooth Low Energy (BLE) and virtual BLE, which means it has location services and internet of things (IoT) connectivity without using physical beacons. Therefore, it is easier to manage a variety of connected devices using the Mist network as a cloud solution for wireless LAN and LAN and IoT or BLE devices.

AI-Driven Network Assurance & Automation

Mist is able to continuously monitor the network health of the APs, switches, or other network devices for anomalies or issues such as interference, congestion, or connectivity. In most cases, the system is also capable of correcting the problem. Administrators can set the expectations for the services (SLEs) in parameters such as connection time, throughput, coverage, latency, and so on. The Mist tracks the SLEs and notifies if there is any deviation in the parameters.

Centralized Management & Visibility

Cloud portal provided by the Mist which can be used set up network configurations, create WLAN templates, lay down policies, manage the profiles of devices, and manage multiple sites from a central location. It provides you with complete visibility of your network. The extent of your network reaches from per-client levels to overall network behaviour for the entire site. The capabilities of Marvis extend to the organizational, site as well as client levels. Such visibility enables various functions like capacity management, performance tracking, security, and troubleshooting. All this ensures network management becomes simpler.

Scalability, Flexibility & Reduced Operational Overhead

Mist is cloud-managed and infrastructure-agnostic scaling up adding more APs or switches, onboarding new sites, supporting more clients becomes easier and faster. No need to deploy additional controllers or do manual per-device configuration. AI-driven automation and insight drastically reduce manual tasks. Routine tasks like radio-planning, interference mitigation, configuration updates, network tuning, troubleshooting are simplified or automated lowering operational cost and saving time.

Configuration of Devices

In this work, we began by configuring the core network devices that provide wired and wireless connectivity. The goal was to establish a secured internal LAN, enable DHCP services, enforce security zones and policies, and support wireless network deployment using Mist AI.

The initial configuration involved the SRX300 firewall and EX2300 switch, followed by the setup of Mist APs for Wi-Fi network optimization.

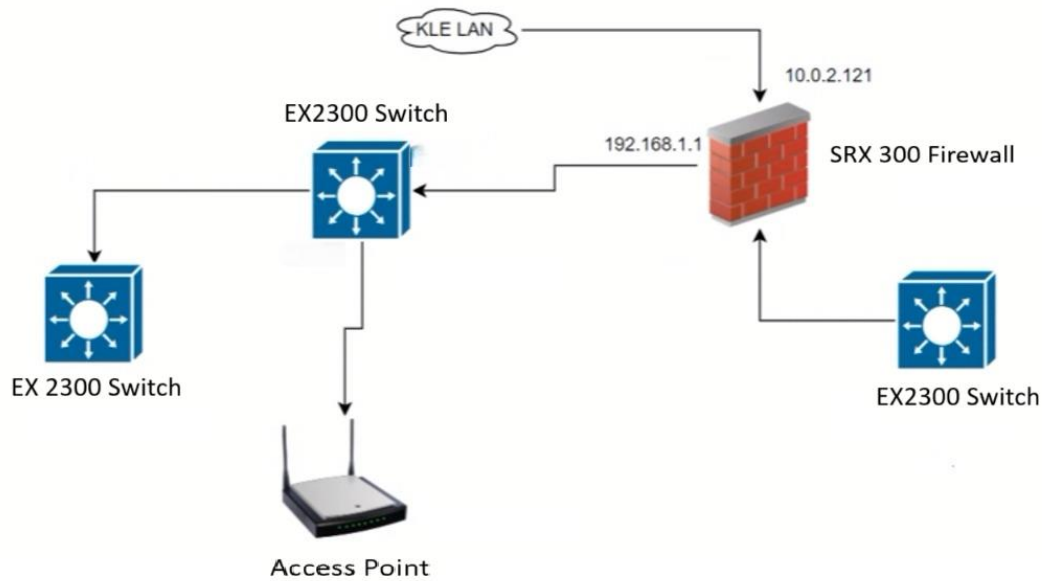


Fig 1: Architecture of Our Setup

Devices Configured:

Device	Role
SRX300	Firewall, NAT, Routing
EX2300	Layer-2 Access switch
Mist AP32 & AP63	Wireless coverage deployment

CONFIGURATION OF SRX 300 AS FIREWALL

Layer 3 Interface IP Assignment:

```

set interfaces ge-0/0/2 unit 0 family inet address 172.20.99.210/24
set interfaces ge-0/0/5 unit 0 family inet address 192.168.10.1/24
  
```

This assigns **IP addresses** to the physical interfaces so the firewall can **route traffic** between networks.

Defines network into 2 segments:

Interface	Assigned Zone	Purpose
ge-0/0/2	Untrust (WAN)	Connects to ISP / Internet
ge-0/0/5	Trust (LAN)	Connects to internal network / clients



ge-0/0/2 communicates with the upstream router using IP 172.20.99.210/24

ge-0/0/5 becomes the default gateway for LAN clients using 192.168.10.1/24

DHCP Server and Address Pool Configuration:

```
set system services dhcp pool 192.168.10.0/24 address-range low 192.168.10.10
```

```
set system services dhcp pool 192.168.10.0/24 address-range high 192.168.10.254
```

```
set system services dhcp pool 192.168.10.0/24 name-server 8.8.8.8
```

```
set system services dhcp pool 192.168.10.0/24 router 192.168.10.1
```

This enables :

a) automatic IP allocation for LAN devices.

- b) DNS and Gateway configuration is sent along with IP.
- c) Prevents manual IP assignment and reduces errors.

Source NAT (SNAT) Configuration:

```
set security nat source rule-set trust-to-untrust from zone trust
set security nat source rule-set trust-to-untrust to zone untrust
set security nat source rule-set trust-to-untrust rule src-nat match source-address 0.0.0.0/0
set security nat source rule-set trust-to-untrust rule src-nat then source-nat interface
set security nat source rule-set trust-to-untrust rule source-nat-rule match source-address 0.0.0.0/0
set security nat source rule-set trust-to-untrust rule source-nat-rule then source-nat interface
```

This enables:

- a) Internal 192.168.x.x clients to access the internet.
- b) Converts private IPs to Public interface IP.
- c) Adds protection by hiding internal IP's.

Security Policy Configuration:

```
set security policies from-zone trust to-zone untrust policy allow-all match source-address any
set security policies from-zone trust to-zone untrust policy allow-all match destination-address any
set security policies from-zone trust to-zone untrust policy allow-all match application any
set security policies from-zone trust to-zone untrust policy allow-all then permit
set security policies from-zone trust to-zone untrust policy trust-to-untrust match source-address any
set security policies from-zone trust to-zone untrust policy trust-to-untrust match destination-address any
set security policies from-zone trust to-zone untrust policy trust-to-untrust match application any
set security policies from-zone trust to-zone untrust policy trust-to-untrust then permit
set security policies from-zone untrust to-zone trust policy block-all match source-address any
set security policies from-zone untrust to-zone trust policy block-all match destination-address any
set security policies from-zone untrust to-zone trust policy block-all match application any
set security policies from-zone untrust to-zone trust policy block-all then deny
set security policies from-zone trust to-zone trust policy trust-to-trust match source-address any
```

```
set security policies from-zone trust to-zone trust policy trust-to-trust match destination-address any
set security policies from-zone trust to-zone trust policy trust-to-trust match application any
set security policies from-zone trust to-zone trust policy trust-to-trust then permit
```

These commands define how traffic flows between different security zones on the SRX300 firewall.

Policy Scope	Result	Reason
Trust to Untrust	Allowed	Users need Internet access
Untrust to Trust	Blocked	Security against outside attackers
Trust to Trust	Allowed	Local communication required

Security Zone Assignment and Interface Binding:

```
set security zones security-zone trust interfaces ge-0/0/5.0 host-inbound-traffic
system-services dhcp
set security zones security-zone untrust interfaces ge-0/0/2.0
```

These steps help to assign interfaces and assigning interfaces to zones helps the SRX300:

- Apply correct security policies
- Identify which traffic is internal and which is external
- Prevent accidental exposure of internal devices to the public network
- Enable DHCP only on trusted internal interface

Default Static Route Configuration:

```
set routing-options static route 0.0.0.0/0 next-hop 172.20.99.1
```

A default route is being set on the SRX300.

Purpose of this configuration:

- Send traffic to outside networks.

- 2) Define a WAN exit path.
- 3) Essential for NAT and Policy flow.



Configuration of EX2300 as Router

Access Port (Layer 2) Configuration

```
set interfaces ge-0/0/4 unit 0 family ethernet-switching interface-mode access
```

```
set interfaces ge-0/0/11 unit 0 family ethernet-switching interface-mode access
```


Two interfaces ge-0/0/4 and ge-0/0/11 on the EX2300 switch are configured as:

- 1) Access Ports to Layer-2 ports that connect directly to end-user devices
- 2) This allows users to communicate within the same LAN.
- 3) The switch does not assign IPs to clients as DHCP is done by SRX300.



Default Static Route Configuration

```
set routing-options static route 0.0.0.0/0 next-hop 192.168.1.1
```

A default route is configured on EX2300 with next-hop 192.168.1.1. This informs the switch where to send traffic that is not meant for its local network.

Purpose:

- 1) Provide an exit path for non-local traffic.
- 2) Route management traffic when required.

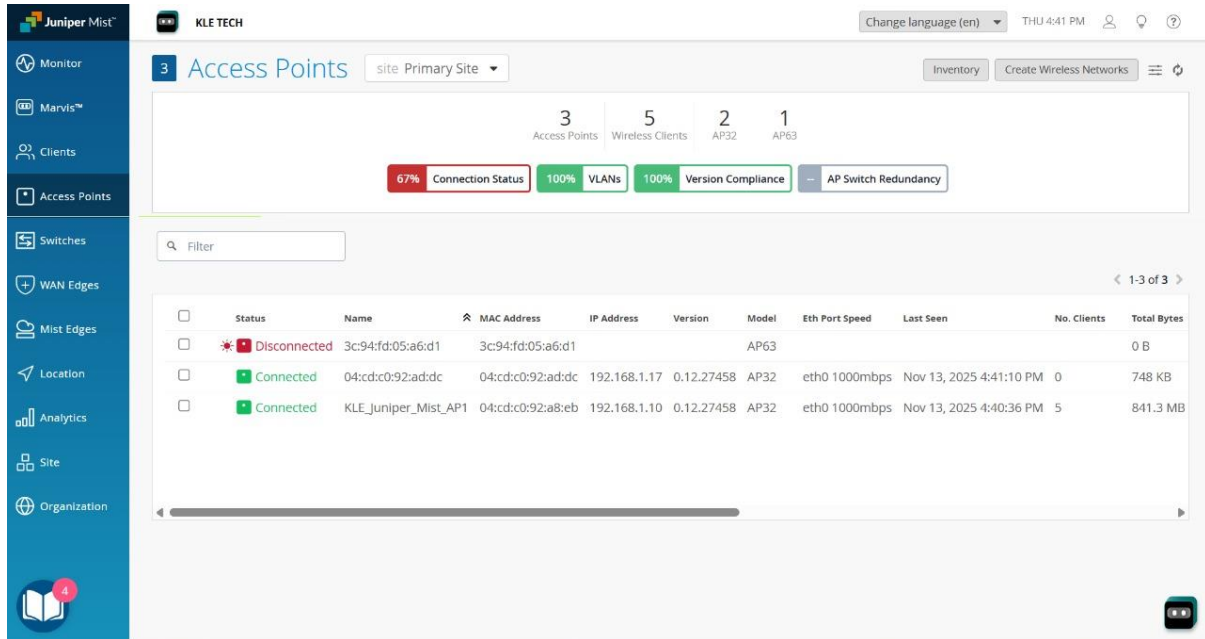


Accessing Web Dashboard of Juniper Mist

To set up this network, the physical and logical connections were established based on the trusted and untrusted zones of the SRX300 firewall. The Internet/WAN uplink was connected to the ge-0/0/2 interface, which is configured as the Untrust zone, while the internal LAN network was connected to the ge-0/0/5 interface, assigned to the Trust zone. The configuration procedure began by assigning static IP addresses to these interfaces and enabling a DHCP server on the SRX300 to automatically allocate IP addresses to internal devices. Following this, Source NAT (SNAT) was configured to translate private internal IPs when accessing the Internet.

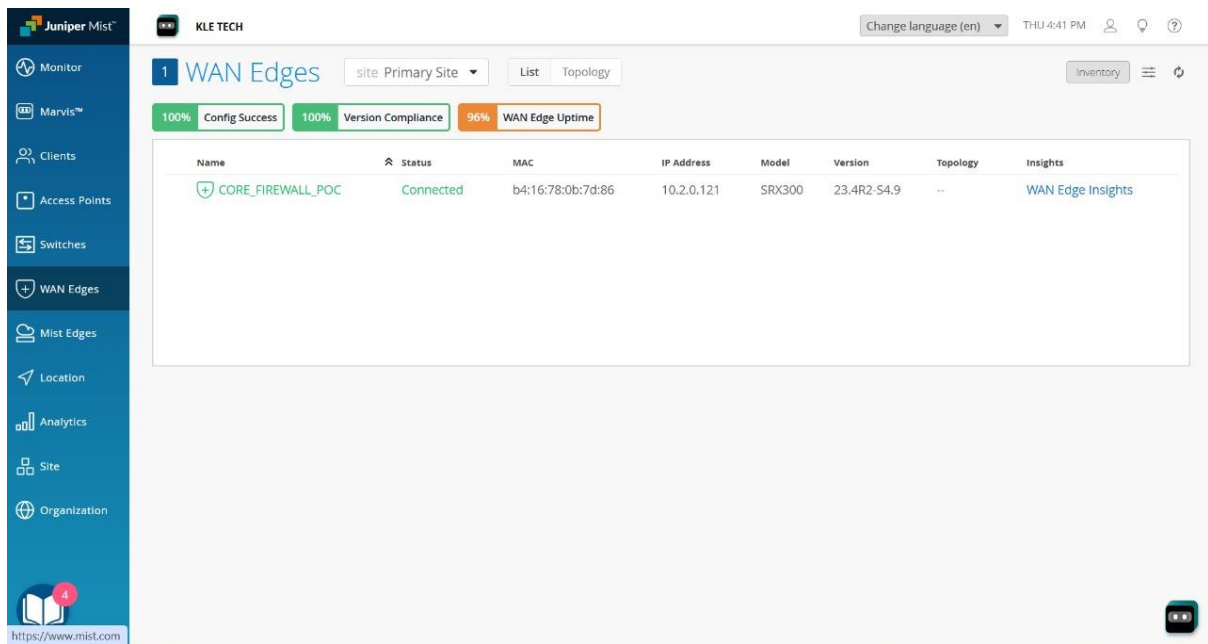
Security policies were implemented to allow outbound traffic from the Trust zone to the Untrust zone while blocking any unauthorized inbound access from the Internet. Additionally, the interfaces were bound to their respective security zones for proper enforcement of firewall rules.

To complete the routing setup, a default static route was added on the SRX300, providing a clear path for traffic destined for external networks. Lastly, the EX2300 switch ports (ge-0/0/4 and ge-0/0/11) were configured as Layer-2 access ports, enabling end-user devices to connect to the Local Area Network through the switch.



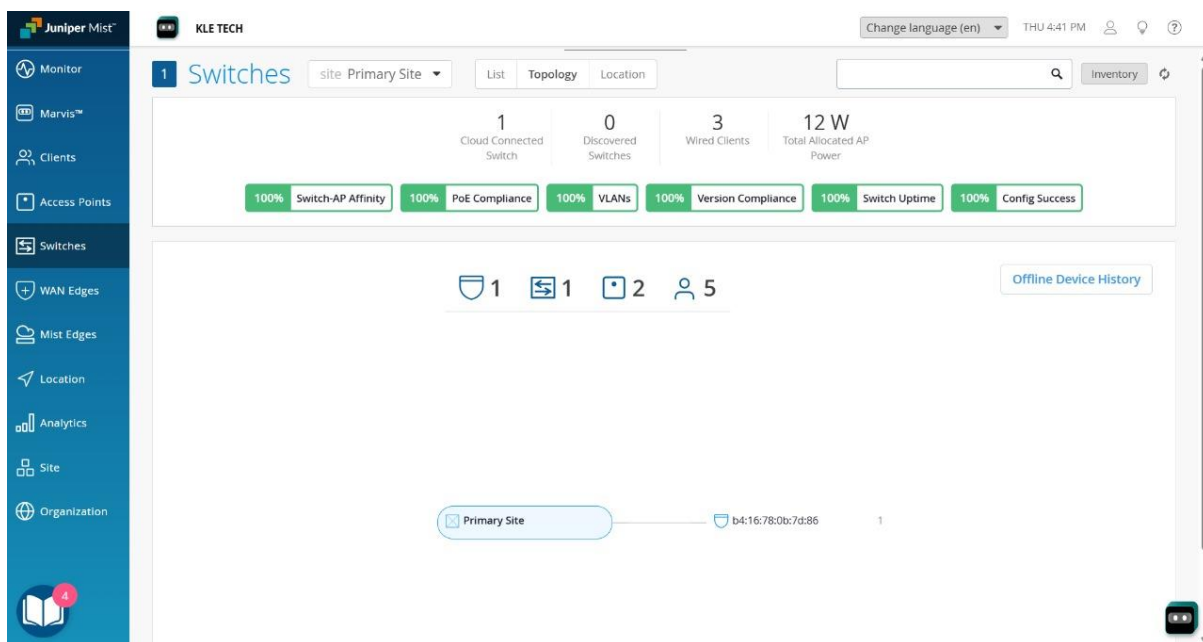
The screenshot shows the Juniper Mist web interface for the 'Access Points' section. The top navigation bar includes 'Monitor', 'Marvis™', 'Clients', 'Access Points', 'Switches', 'WAN Edges', 'Mist Edges', 'Location', 'Analytics', 'Site', and 'Organization'. The 'Access Points' section is active, showing a summary of 3 Access Points, 5 Wireless Clients, 2 AP32, and 1 AP63. Below this, there are status indicators: 67% Connection Status, 100% VLANs, 100% Version Compliance, and AP Switch Redundancy. A table lists the access points with columns for Status, Name, MAC Address, IP Address, Version, Model, Eth Port Speed, Last Seen, No. Clients, and Total Bytes.

Status	Name	MAC Address	IP Address	Version	Model	Eth Port Speed	Last Seen	No. Clients	Total Bytes
Disconnected	3c94:fd:05:a6:d1	3c94:fd:05:a6:d1			AP63				0 B
Connected	04:cd:c0:92:ad:dc	04:cd:c0:92:ad:dc	192.168.1.17	0.12.27458	AP32	eth0 1000mbps	Nov 13, 2025 4:41:10 PM	0	748 KB
Connected	KLE_Juniper_Mist_AP1	04:cd:c0:92:a8:eb	192.168.1.10	0.12.27458	AP32	eth0 1000mbps	Nov 13, 2025 4:40:36 PM	5	841.3 MB



The screenshot shows the Juniper Mist web interface for the 'WAN Edges' section. The top navigation bar is the same as the previous screenshot. The 'WAN Edges' section is active, showing a summary of 100% Config Success, 100% Version Compliance, and 96% WAN Edge Uptime. Below this, a table lists the WAN edges with columns for Name, Status, MAC, IP Address, Model, Version, Topology, and Insights.

Name	Status	MAC	IP Address	Model	Version	Topology	Insights
CORE_FIREWALL_POC	Connected	b4:16:78:0b:7d:86	10.2.0.121	SRX300	23.4R2-S4.9	--	WAN Edge Insights



Juniper Mist KLE TECH

Change language (en) THU 4:41 PM

5 WiFi Clients

site Primary Site Live Guest Marvis

5 Wireless Clients 1 2.4 GHz 4 5 GHz 2 802.11ac 3 802.11ax 5 KLE_JUNIPER

Filter

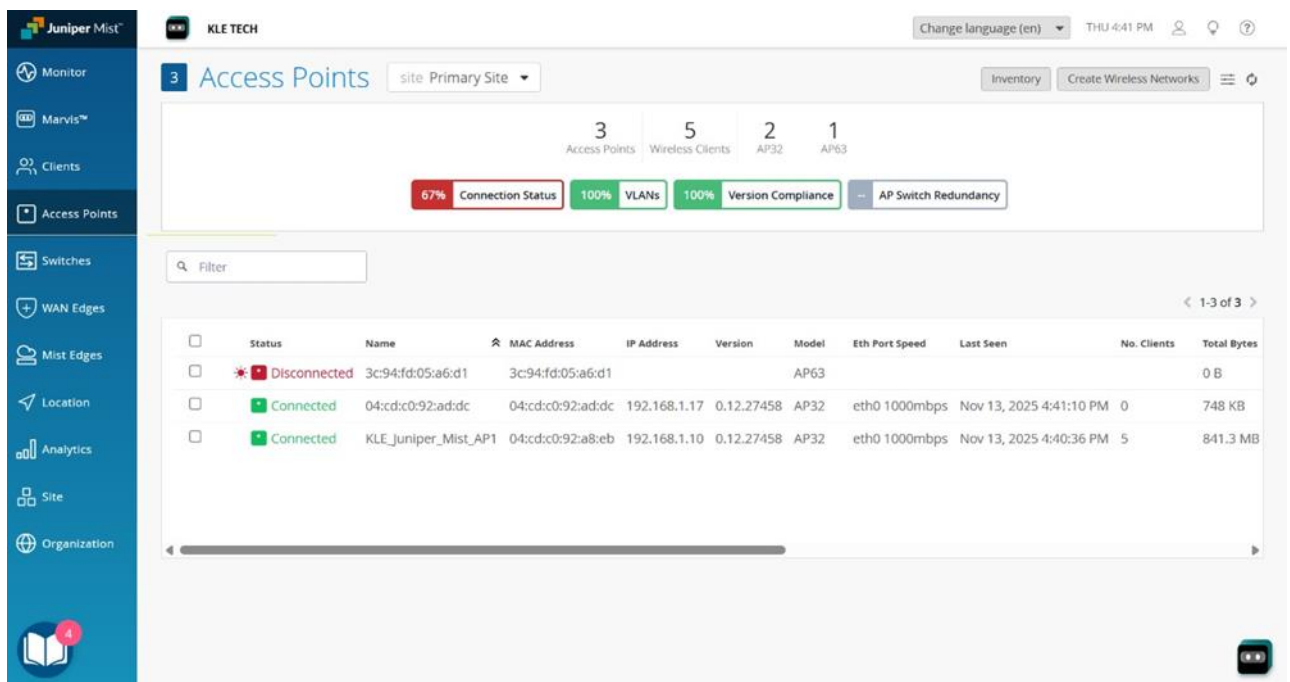
< 1-5 of 5 >

User	IPv4 Address	MAC Address	Device Type	AP Name	SSID	Pre-Shared Key	Insights
☐ iPhone	192.168.1.12	6a:99:17:20:08:4a	IOS	KLE_Juniper_Mist_AP1	KLE_JUNIPER		Client Insights
☐ Madhav	192.168.1.16	3c:e9:f7:d2:95:97	Intel Corporate	KLE_Juniper_Mist_AP1	KLE_JUNIPER		Client Insights
☐ motorola-edge-60-fusion	192.168.1.14	4a:58:1c:0a:74:d0	Motorola Edge	KLE_Juniper_Mist_AP1	KLE_JUNIPER		Client Insights
☐ Nagarjuna	192.168.1.13	98:2c:bc:25:8e:0b	Intel Corporate	KLE_Juniper_Mist_AP1	KLE_JUNIPER		Client Insights
☐ V2420	192.168.1.15	6e:c7:d1:f5:24:28	Unknown	KLE_Juniper_Mist_AP1	KLE_JUNIPER		Client Insights

Radio Resource Management with Mist AI

Objective: Deploy and optimize wireless coverage using Mist APs and cloud-based management.

- 1) Deploy the AP32 (Indoor) and AP63 (Outdoor) access points to provide seamless coverage across locations.



- 2) Use Mist Cloud GUI to configure SSIDs, enable band steering (2.4 GHz/5 GHz), and analyse user density

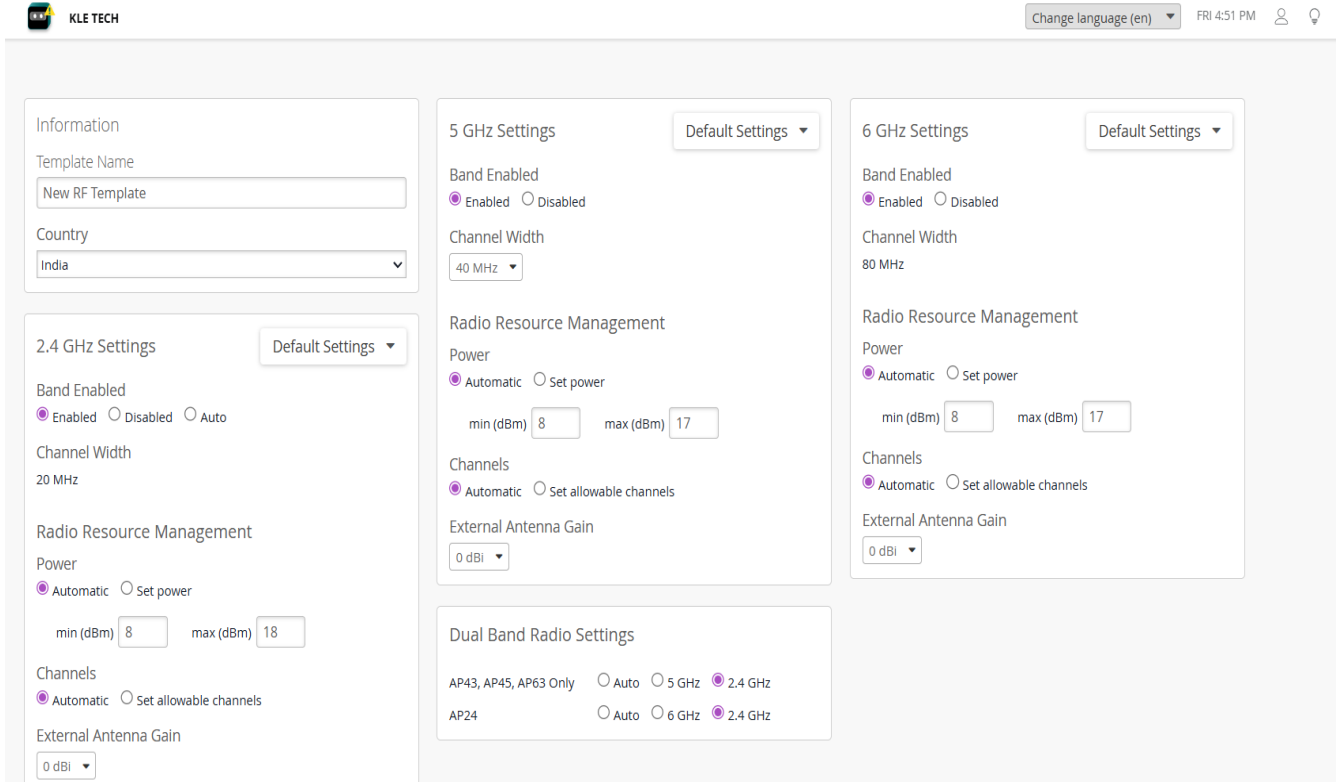
Wireless networks (SSIDs) are created and managed through the Mist Cloud portal.

This allows easy setup of different networks like:

- Corporate Wi-Fi
- Guest Wi-Fi

- IoT network

All settings are centrally controlled and pushed to APs automatically.

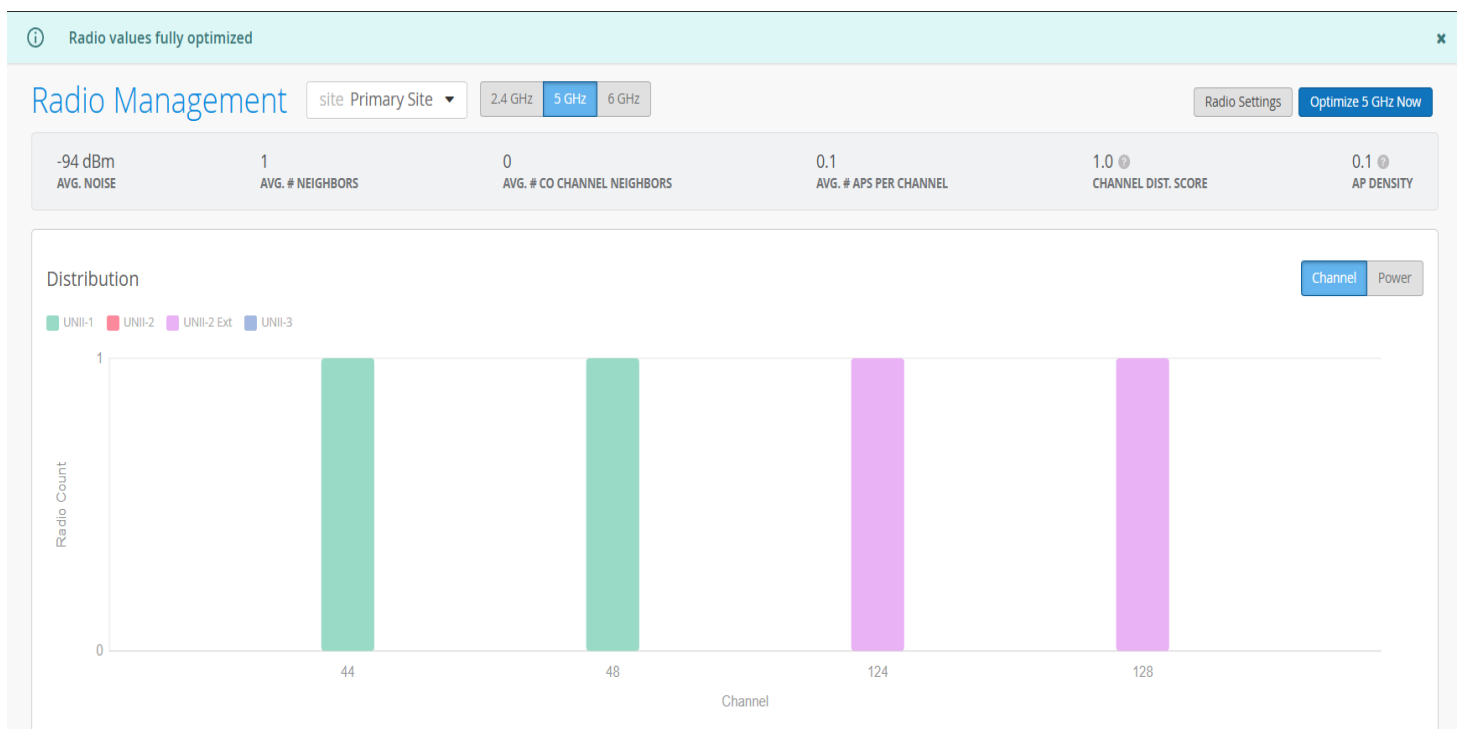
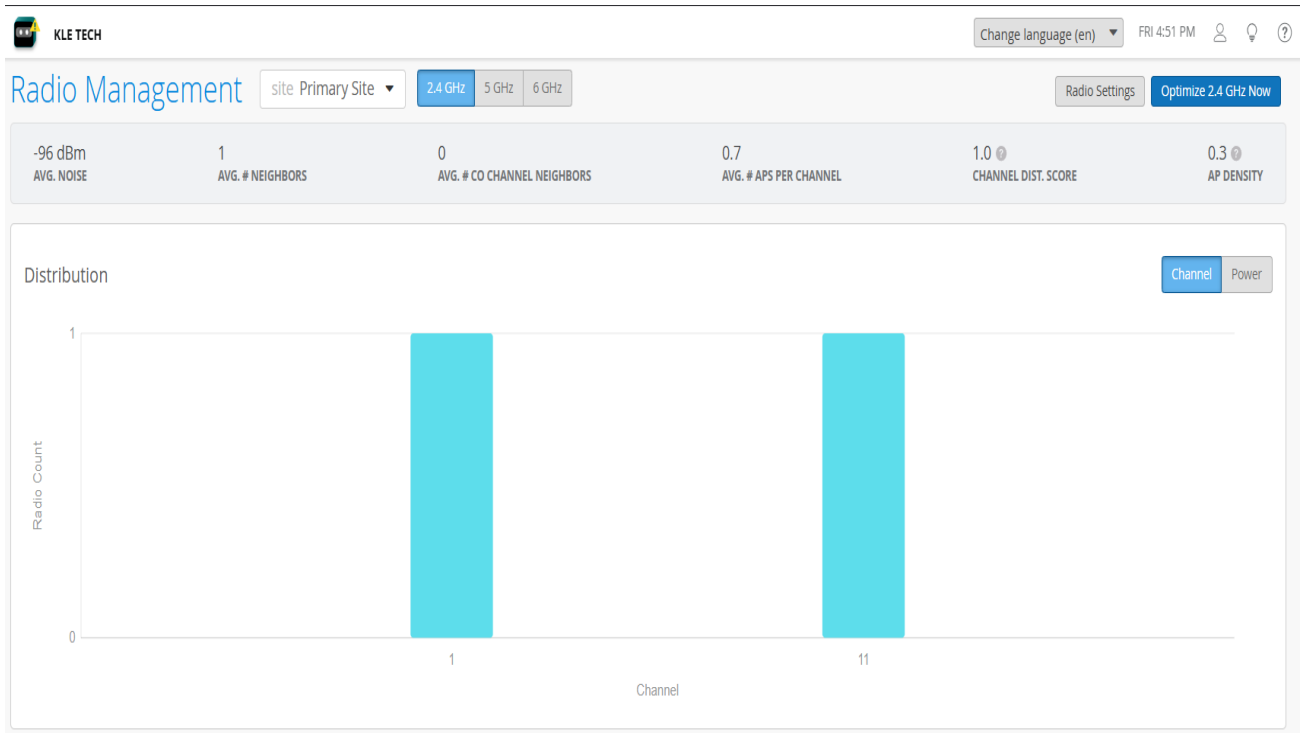


The screenshot displays the KLE TECH IoT network configuration interface. At the top, there is a header with the KLE TECH logo, a language selector (Change language (en)), and the time (FRI 4:51 PM). The main content area is divided into several sections:

- Information:** Includes fields for Template Name (New RF Template) and Country (India).
- 2.4 GHz Settings:**
 - Band Enabled: ☒ Enabled, ☐ Disabled, ☐ Auto
 - Channel Width: 20 MHz
 - Radio Resource Management:
 - Power: ☒ Automatic, ☐ Set power. min (dBm) 8, max (dBm) 18
 - Channels: ☒ Automatic, ☐ Set allowable channels
 - External Antenna Gain: 0 dBi
- 5 GHz Settings:**
 - Band Enabled: ☒ Enabled, ☐ Disabled
 - Channel Width: 40 MHz
 - Radio Resource Management:
 - Power: ☒ Automatic, ☐ Set power. min (dBm) 8, max (dBm) 17
 - Channels: ☒ Automatic, ☐ Set allowable channels
 - External Antenna Gain: 0 dBi
- 6 GHz Settings:**
 - Band Enabled: ☒ Enabled, ☐ Disabled
 - Channel Width: 80 MHz
 - Radio Resource Management:
 - Power: ☒ Automatic, ☐ Set power. min (dBm) 8, max (dBm) 17
 - Channels: ☒ Automatic, ☐ Set allowable channels
 - External Antenna Gain: 0 dBi
- Dual Band Radio Settings:**
 - AP43, AP45, AP63 Only: ☐ Auto, ☐ 5 GHz, ☒ 2.4 GHz
 - AP24: ☐ Auto, ☐ 6 GHz, ☒ 2.4 GHz

3) Optimize performance through Mist's AI-driven Radio Resource Management (RRM).

Band steering improves performance by intelligently pushing capable devices to the faster 5 GHz band, reducing congestion on 2.4 GHz. This helps users experience faster speeds and better connectivity.

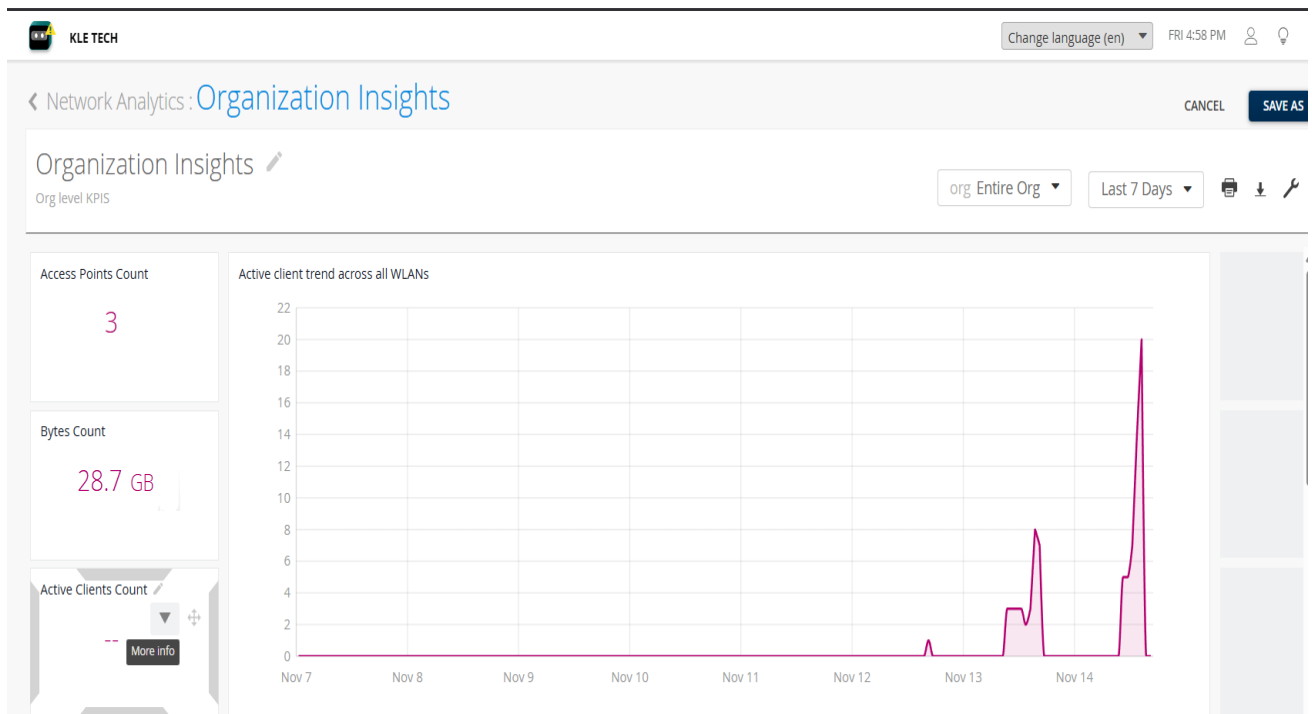


4) Use Mist AI Insights to troubleshoot connectivity, latency, or interference issues.

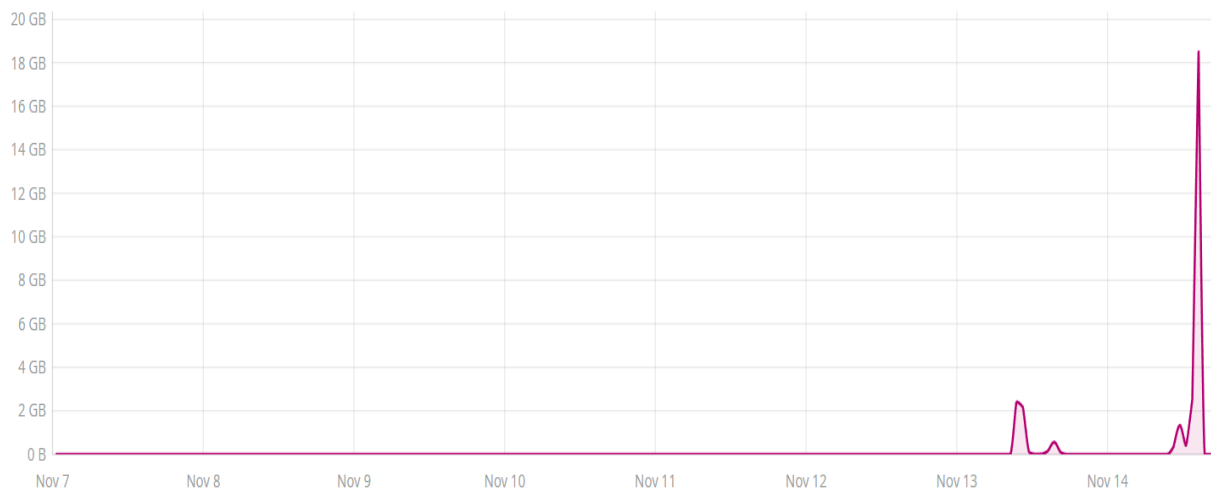
Mist AI analyses network issues such as:

- Client connection failures
- Latency or low throughput
- Coverage or RF interference problems

It provides actionable recommendations to improve the network and troubleshoot faster.

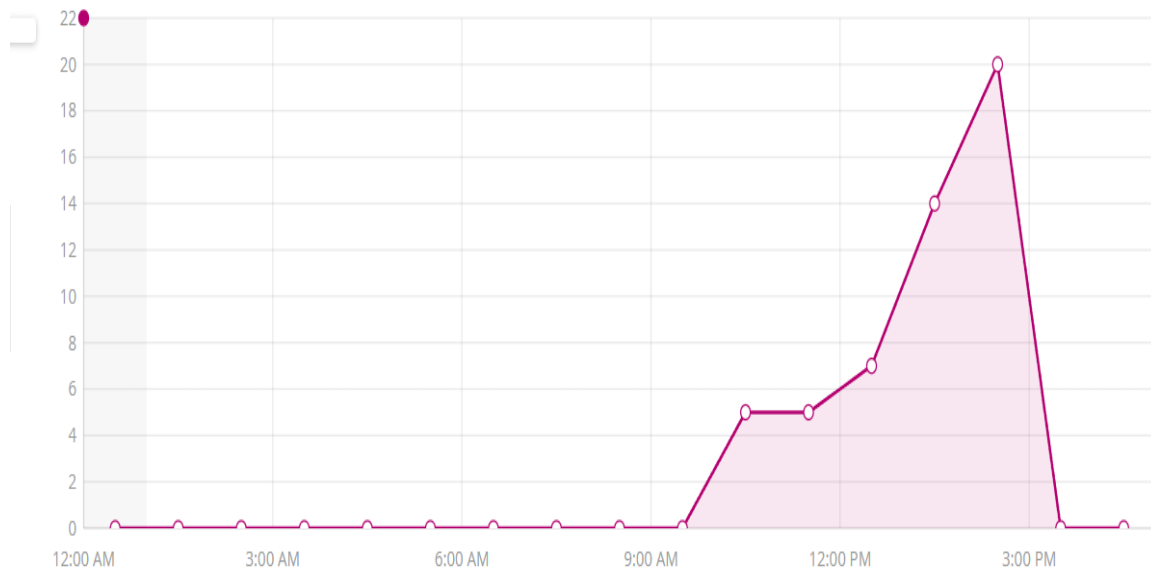


Traffic Utilization across all WLANs

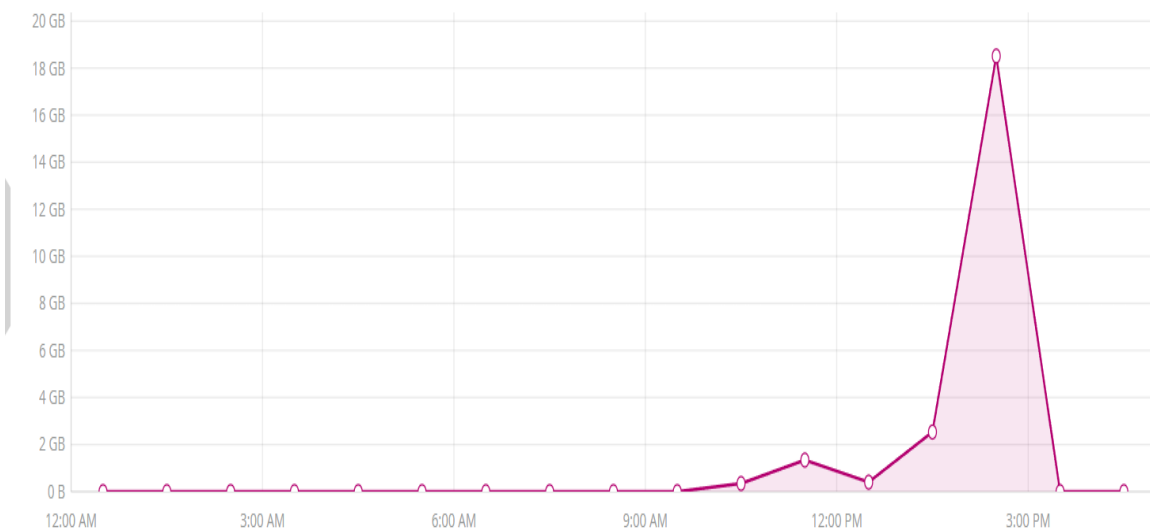




Active client trend across all WLANs



Traffic Utilization across all WLANs



Task 2: Fetching Access Point data using API's

In large-scale enterprise wireless networks, continuous monitoring of Access Points (APs), connected clients, and network health indicators is critical for ensuring reliable connectivity and optimal performance. Manual monitoring through graphical dashboards is often insufficient for real-time analysis, automation, and integration with external systems.

To address this, Juniper Mist provides a comprehensive set of RESTful APIs that allow programmatic access to real-time telemetry, device statistics. These APIs enable automated polling of network data, facilitating proactive monitoring, troubleshooting, and performance evaluation.

In this task, a Mist polling client is implemented using Python to periodically retrieve:

- Live client statistics across the site
- Real-time device (AP) operational statistics
- Site-level AI insights generated by the Mist platform

1) API Authentication Mechanism

Authorization: Token <MIST_API_TOKEN>
Content-Type: application/json

All Mist APIs require **token-based authentication**, ensuring secure access to organizational and site-level data.

1. The API token must have sufficient read privileges for the organization and site.
2. Requests are transmitted securely over HTTPS.
3. Authentication headers are included in every API call to the Mist-cloud.

2) Site Clients Statistics API

GET /api/v1/sites/{site_id}/stats/clients

This API retrieves **real-time statistics of all active wireless clients** connected across the site. Each record corresponds to a single client session and includes detailed connection and traffic information.

Key Parameters Retrieved

1. Client MAC address
2. Assigned IP address
3. Associated AP MAC and AP name
4. SSID used for connection
5. RSSI and SNR values
6. Data usage statistics (TX and RX bytes)
7. Client uptime and connection duration

3) Site Devices Statistics API

GET /api/v1/sites/{site_id}/stats/devices

This API provides **live operational statistics for all network devices** within a site, including Access Points, switches, and gateways.

Key Parameters Retrieved (AP-device)

1. Device name and MAC address
2. Device model (e.g., AP32)
3. Operational status (connected/disconnected)
4. IP address
5. Uptime
6. CPU and memory usage
7. Radio-level statistics (channel, noise floor, utilization)
8. LLDP information indicating switch port connectivity

4) Site Insights API

GET /api/v1/sites/{site_id}/insights/site/{site_id}/stats

This API returns **insights and anomalies** detected by the Mist platform at the site level. These insights are generated using Mist's machine learning models that continuously analyse telemetry data.

Types of Insights Returned

1. Coverage-related issues
2. Capacity and congestion alerts
3. Roaming anomalies
4. Interference detection
5. Service degradation events

5) Radio Statistics API

GET /api/v1/sites/{site_id}/stats/radios

The AP Radio Statistics API provides real-time telemetry for each radio on an access point, including operating channel, noise floor, channel utilization, and transmit power. This information is useful for RF optimization by identifying congested or noisy channels and supporting channel planning decisions. It also helps in detecting interference that can degrade wireless performance.

6) Output:

```
=== MIST AP MONITOR: 12:28:58 ===  
  
[ AP STATUS: KLE_Juniper_Mist_AP1 ]  
Status:    CONNECTED  
Model:     AP32  
IP Addr:   192.168.137.252  
Uptime:    0.75 hours  
LLDP Port:  
  
[ RADIO INSIGHTS ]  
2.4GHz: Ch 1 | Noise: -91 dBm | Util: 24%  
5GHz:   Ch 116 | Noise: -94 dBm | Util: 1%  
  
[ CONNECTED CLIENTS (5) ]  
MAC Address | IP Address | RSSI | Proto | Tx Bytes | Rx Bytes  
-----  
5605792b85ef | 192.168.137.129 | -23 | n | 11.4MB | 1.9MB  
505a65b1088f | 192.168.137.121 | -48 | ax | 51.1MB | 14.9MB  
0af0955676f4 | 192.168.137.207 | -59 | n | 17.2MB | 2.6MB  
aee8969c6786 | 192.168.137.206 | -41 | ac | 129.7KB | 104.7KB  
3eed8162e9e0 | 192.168.137.169 | -64 | ac | 505.8KB | 793.6KB
```



```
--- Clients @ 2026-01-02T07:00:14.832107Z
[
  {
    "mac": "5605792b85ef",
    "site_id": "4c40f225-082f-48b9-a9ab-563e157b5a4e",
    "assoc_time": 1767334475,
    "family": "OnePlus",
    "model": "",
    "os": "Android 15",
    "manufacture": "OnePlus Technology (Shenzhen) Co.",
    "bssid": "04cdc0ac8681",
    "hostname": "OnePlus-Nord-CE-3-Lite-5G",
    "ip": "192.168.137.129",
    "ap_mac": "04cdc092a8eb",
    "ap_id": "000000000-0000-0000-1000-04cdc092a8eb",
    "last_seen": 1767337193,
    "uptime": 2718,
    "ssid": "KLE_JUNIPER",
    "wlan_id": "8edbe7aa-920e-4298-afc7-4e91c9b3a1c3",
    "dual_band": false,
    "is_guest": false,
    "key_mgmt": "WPA2-PSK-FT/CCMP",
    "group": "",
    "band": "24",
    "channel": 1,
    "vlan_id": "1",
    "proto": "n",
    "rssi": -27,
    "snr": 62,

    "channel": 1,
    "vlan_id": "1",
    "proto": "n",
    "rssi": -27,
    "snr": 62,
    "idle_time": 6.0,
    "tx_rate": 72.2,
    "rx_rate": 1.0,
    "tx_pkts": 12627,
    "rx_pkts": 9797,
    "tx_bytes": 11938311,
    "rx_bytes": 1969321,
    "tx_retries": 358,
    "rx_retries": 605,
    "tx_bps": 0,
    "rx_bps": 0,
    "psk_id": "",
    "_ttl": 282,
    "_id": "5605792b85ef",
    "annotation": "unknown"
  }
]
```



```
{
  "mac": "505a65b1088f",
  "site_id": "4c40f225-082f-48b9-a9ab-563e157b5a4e",
  "assoc_time": 1767335384,
  "family": "",
  "model": "",
  "os": "Windows 10/11",
  "manufacture": "AzureWave Technology Inc.",
  "bssid": "04cdc0ac8681",
  "hostname": "Abhishek_M",
  "ip": "192.168.137.121",
  "ap_mac": "04cdc092a8eb",
  "ap_id": "00000000-0000-0000-1000-04cdc092a8eb",
  "last_seen": 1767337193,
  "uptime": 1809,
  "ssid": "KLE_JUNIPER",
  "wlan_id": "8edbe7aa-920e-4298-afc7-4e91c9b3a1c3",
  "dual_band": false,
  "is_guest": false,
  "key_mgmt": "WPA2-PSK/CCMP",
  "group": "",
  "band": "24",
  "channel": 1,
  "vlan_id": "1",
  "proto": "ax",
  "rssi": -49,
  "snr": 40,
  "idle_time": 26.0,
  "tx_rate": 258.0,
  "rx_rate": 1.0,
}
```

```
{  
  "mac": "0af0955676f4",  
  "site_id": "4c40f225-082f-48b9-a9ab-563e157b5a4e",  
  "assoc_time": 1767336027,  
  "family": "iPhone",  
  "model": "",  
  "os": "17.2",  
  "manufacture": "Apple",  
  "bssid": "04cdc0ac8681",  
  "ip": "192.168.137.207",  
  "ap_mac": "04cdc092a8eb",  
  "ap_id": "000000000-0000-0000-1000-04cdc092a8eb",  
  "last_seen": 1767337193,  
  "uptime": 1166,  
  "ssid": "KLE_JUNIPER",  
  "wlan_id": "8edbe7aa-920e-4298-afc7-4e91c9b3a1c3",  
  "dual_band": false,  
  "is_guest": false,  
  "key_mgmt": "WPA2-PSK-FT/CCMP",  
  "group": "",  
  "band": "24",  
  "channel": 1,  
  "vlan_id": "1",  
  "proto": "n",  
  "rssi": -58,  
  "snr": 31,  
  "idle_time": 4.0,  
  "tx_rate": 65.0,  
  "rx_rate": 1.0,  
}
```



```
"dual_band": false,  
"is_guest": false,  
"key_mgmt": "WPA2-PSK-FT/CCMP",  
"group": "",  
"band": "24",  
"channel": 1,  
"vlan_id": "1",  
"proto": "n",  
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"tx_rate": 65.0,  
"rx_rate": 1.0,  
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"tx_bytes": 18120996,  
"rx_bytes": 2792896,  
"tx_retries": 7063,  
"rx_retries": 3192,  
"tx_bps": 0,  
"rx_bps": 0,  
"psk_id": "",  
"_ttl": 282,  
"_id": "0af0955676f4",  
"annotation": "unknown"  
},  
{  
  "mac": "aee8969c6786",  
  "site_id": "4c40f225-082f-48b9-a9ab-563e157b5a4e",  
  "assoc_time": 1767336681,
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    "netmask": "255.255.255.0",
    "dhcp_server": "192.168.1.1"
  },
  "name": "KLE_Juniper_Mist_AP2",
  "inactive_wired_vlans": [],
  "tx_pkts": 160897,
  "lldp_stats": {
    "eth0": {
      "system_name": "KLE_Juniper_SW",
      "system_desc": "Juniper Networks, Inc. ex2300-c-12p Ethernet Switch, kernel
5-2025 Juniper Networks, Inc.",
      "mgmt_addr": "c0-df-ed-b7-f5-86",
      "mgmt_addrs": [
        "c0-df-ed-b7-f5-86"
      ],
      "port_desc": "ge-0/0/4",
      "port_id": "ge-0/0/4",
      "ap_port_name": "",
      "chassis_id": "c0:d
--- Site Insights @ 2026-01-02T09:44:00.830650Z
{
  "start": 1767261600,
  "end": 1767348000,
  "interval": 3600,
  "limit": 168,
  "page": 1
}
```

Conclusion

Through this implementation, we gained practical experience in designing and deploying a secure network using Juniper SRX300 and EX2300 devices. By configuring essential networking components such as routing, DHCP, NAT, and security policies, we successfully established reliable LAN-to-WAN connectivity while enforcing proper access control and network security for devices.

After building a strong wired network foundation, we deployed Mist-managed wireless Access Points and leveraged the Mist Cloud platform to monitor and optimize Wi-Fi performance. Using Mist APIs, we were able to programmatically retrieve real-time Access Point statistics, client information, and AI-driven insights, enabling automated monitoring and data-driven analysis of the wireless network. This work also provided valuable insight into AI-driven networking concepts, such as Radio Resource Management (RRM) and Mist Insights, which dynamically optimize channel selection, power levels, and detect performance anomalies. These capabilities significantly simplify troubleshooting and reduce manual intervention by proactively identifying coverage, capacity, and interference issues.

Overall, this implementation helped us understand how Juniper's secure networking devices and Mist AI platform work together to deliver a secure, scalable, and intelligent enterprise networking solution. The project bridged theoretical networking concepts with real-world deployment and automation, strengthening our understanding of modern cloud-managed and AI-assisted network infrastructures.